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title: "SQL Report load_data.sql"
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date: "2026-03-17"
output: pdf
Title: CS 504 project stage 2 load_data.sql
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-- To load data into local database, from the same directory, or use the full path.
-- psql -h 127.0.0.1 -p 5432 -U postgres -d cs-504 -f load_data.sql

-- =====
-- 1. Drop tables (optional, for a clean reload)
-- =====
DROP TABLE IF EXISTS authors          CASCADE;
DROP TABLE IF EXISTS catalog_tbl      CASCADE;
DROP TABLE IF EXISTS genre            CASCADE;
DROP TABLE IF EXISTS member_tbl       CASCADE;
DROP TABLE IF EXISTS staff            CASCADE;
DROP TABLE IF EXISTS material          CASCADE;
DROP TABLE IF EXISTS authorship        CASCADE;
DROP TABLE IF EXISTS borrow           CASCADE;
-- =====
-- 2. Create tables
--   table_name: attributes
--   authors:author_id,name,nationality
--   authorship:authorship_id,author_id,material_id
--   borrow:borrow_id,material_id,member_id,staff_id,borrow_date,due_date,return_date
--   catalog:catalog_id,name,location
--   genre:genre_id,name,description
--   material:material_id,title,publication_date,catalog_id,genre_id
--   member_tbl:member_id,name,contact_info,join_date
--   staff:staff_id,name,contact_info,job_title,hire_date
-- =====

-- Notice: the first 5 tables have no dependencies
CREATE TABLE authors (
  author_id  INTEGER PRIMARY KEY,
  name       TEXT NOT NULL,
  nationality TEXT
);

CREATE TABLE catalog_tbl (
  catalog_id  INTEGER PRIMARY KEY,
  name        TEXT NOT NULL,
  location    TEXT
);

CREATE TABLE genre (
  genre_id    INTEGER PRIMARY KEY,
  name        TEXT NOT NULL,
  description  TEXT
);

CREATE TABLE member_tbl (
  member_id   INTEGER PRIMARY KEY,
  name        TEXT NOT NULL,
  contact_info TEXT,
  join_date   DATE
);

CREATE TABLE staff (

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    staff_id    INTEGER PRIMARY KEY,
    name        TEXT NOT NULL,
    contact_info TEXT,
    job_title   TEXT,
    hire_date   DATE
);

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-- Notice: Material table must be created before authorship and borrow

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CREATE TABLE material (
    material_id    INTEGER PRIMARY KEY,
    title          TEXT NOT NULL,
    publication_date INTEGER,
    catalog_id     INTEGER REFERENCES catalog_tbl(catalog_id),
    genre_id       INTEGER REFERENCES genre(genre_id)
);

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CREATE TABLE authorship (
    authorship_id INTEGER PRIMARY KEY,
    author_id     INTEGER REFERENCES authors(author_id),
    material_id   INTEGER REFERENCES material(material_id)
);

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CREATE TABLE borrow (
    borrow_id    INTEGER PRIMARY KEY,
    material_id  INTEGER REFERENCES material(material_id),
    member_id    INTEGER REFERENCES member_tbl(member_id),
    staff_id     INTEGER REFERENCES staff(staff_id),
    borrow_date  DATE,
    due_date     DATE,
    return_date  DATE
);

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-- =====
-- 3. Load data from CSV files
--   Paths are relative
--   Keep same order as creation
-- =====

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\copy authors FROM './Data/Authors.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', NULL
'', QUOTE '');
\copy catalog_tbl FROM './Data/Catalog.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',',
NULL '', QUOTE '');
\copy genre FROM './Data/Genre.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', NULL '',
QUOTE '');
\copy member_tbl FROM './Data/Member.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', NULL
'', QUOTE '');
\copy staff FROM './Data/Staff.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', NULL '',
QUOTE '');
\copy material FROM './Data/Material.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', NULL
'', QUOTE '');
\copy authorship FROM './Data/Authorship.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',',
NULL '', QUOTE '');
\copy borrow FROM './Data/Borrow.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', NULL '',
QUOTE '');

```

